

Tata Steel AI Hackathon - Comprehensive Interview Questions

1. Can you walk us through the end-to-end architecture of your Industrial AI Assistant?
2. Why didn't you deploy this online or provide a web link? Why run it locally?
3. What makes your idea different from the other teams who used API services like OpenAI or Gemini?
4. Your video was 23 minutes long. Can you give us the 2-minute elevator pitch?
5. What was the hardest technical challenge you faced while building this system?
6. How do you handle processing and storing heavy PDF safety manuals?
7. How would you redesign your database to scale this architecture for all Tata Steel plants globally?
8. How do you prevent the AI from giving dangerous or incorrect safety advice (hallucinations)?
9. Why did you separate the dashboard into Manager, Supervisor, and Engineer portals?
10. How does the Hugging Face Vision Model detect and classify equipment defects from an image?
11. How does the Chatbot connect to the SQLite Inventory Database automatically?
12. What are the strict data security benefits of running local Ollama models compared to cloud APIs?
13. How did you manage system memory when running heavy local LLMs and Vision models on a single machine? (Lazy Loading / Fallbacks)
14. Can you explain how your Retrieval-Augmented Generation (RAG) pipeline works step-by-step?
15. What embedding model did you choose for ChromaDB and why?
16. How does the Supervisor Approval workflow function, and why is human verification necessary?
17. What happens to your application if the Tata Steel factory loses connection to the outside internet? Can it still function?
18. How did you handle the backend data streaming so that the AI response feels real-time instead of freezing the browser?
19. Why did you choose FastAPI over other Python frameworks like Django or Flask for this project?
20. If Tata Steel gave you funding to deploy this into production tomorrow, what is the very first

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component you would upgrade or change?